

Giving the Model Eyes: Ghost Dimensions as an Introspection Prosthetic

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With Apart Research

Abstract

Track 3 asks whether models have privileged access to their own internal states. We report a geometric finding and its characterization. In Qwen3.5-72B, PC1 of the residual stream — carrying 28-67% of activation variance — is excluded from J-space (the verbalizable workspace) at mid-network layers (cosine ≈ 0.003 at L18-L40). We call these “ghost dimensions.”

Across 167 probe snapshots (47 from agentic narrative sessions, 120 from baselines), we find that ghost and workspace probes measure genuinely distinct content: 97.6% of ghost vocabulary tokens never appear in workspace readings. The ghost vocabulary is metacognitive — dominated by tokens about memory itself (`memories`, `/recollection`, `/memories`) — while workspace tokens carry scene-relevant semantics. Ghost-workspace separation varies with context: agentic narrative produces significantly lower ghost cosine (0.099) than isolated recall baselines (0.414, $p < 0.0001$), and ghost and circumplex probes are orthogonal ($r = -0.001$, $p = 0.997$). We also report an introspection prosthetic (GhostReading) that returns ghost content to the agent. The elicitation experiment testing whether agents can use this access was not executed during the sprint; the characterization and probe separation findings stand independently.

1. Introduction

Track 3 asks whether models have privileged access to their own internal states. The standard question is “can the model accurately report its internal states?” (Lindsey 2025). We ask a different question: what does the model process that it *cannot* report — and what is that content?

Ghost dimensions are high-variance processing directions excluded from J-space (Gurnee et al. 2026), the verbalizable workspace. The model performs substantial computation along these axes — the variance is real, the decoded vocabulary is interpretable — but the content never reaches the output pathway. This is not an architectural constraint we imposed; it is what the J-lens reveals about the model’s own geometry.

We built a system that shows the model what its ghost dimensions carry: the GhostReading mechanism records the dominant vocabulary, J-space exclusion cosine, and variance fraction at each retrieval event, returning this to the agent alongside its retrieval result. The model now has access to processing it previously could not report.

Contributions:

1. Characterization of ghost dimensions in Qwen3.5-72B: PC1 carries 28-67% of variance yet is excluded from J-space (cos ≈ 0.003 at mid-network layers).

2. Ghost vocabulary analysis across 167 snapshots: ghost content is metacognitive (tokens about memory itself), workspace content is semantic (scene-relevant), and the two are 97.6% non-overlapping.
3. Context-dependent ghost separation: agentic narrative produces $4\times$ stronger ghost-workspace separation than isolated baselines ($p < 0.0001$).
4. The GhostReading introspection prosthetic: design and implementation. The planned elicitation test (§3.3) was not executed during the sprint.

2. Related Work

The J-lens (Gurnee et al. 2026) identifies J-space — the verbalizable workspace — as approximately 10% of total activation variance. Burns et al. (2023) demonstrate latent knowledge beyond surface outputs via contrast-consistent search. Zou et al. (2023) show that internal directions can be read and steered. Lindsey (2025) finds models detect concept-injected states at $\sim 20\%$ accuracy. The logit lens (nostalgebraist 2020) and tuned lens (Belrose et al. 2023) read intermediate representations via the unembedding matrix.

Gap: Prior introspection work asks whether models can report states that experimenters inject or identify. We ask what the model processes *on its own* that never reaches output, what that content is, and whether it varies with context.

3. Methods

3.1 Ghost Dimension Characterization

[PCA on residual stream at each layer $\rightarrow$ PC1 direction] [Logit lens: $W_U \cdot pc1 \rightarrow$ what vocabulary the dimension encodes (structural markers)] [J-lens: $W_U \cdot J_L \cdot pc1 \rightarrow$ what it contributes to output (flat = ghost)] [Cosine between logit-lens and J-lens probability distributions = ghost exclusion metric] [Validated: 3 null checks (H0_1 centering, H0_2 random baseline, H0_3 permutation)] [Matched-variance null (H1): $n = 200$ random directions drawn at PC1’s variance fraction, per layer. Because J-space is $\sim 10\%$ of variance while PC1 is 28-67%, low cosine may be forced by dimensional accounting alone. H1 counts as supported only if the observed cosine falls below the 5th percentile of this null; otherwise the exclusion is reported as trivial, not empirical.]

3.2 The Ghost Reading (Introspection Prosthetic)

[GhostReading dataclass: `pc1_variance_pct`, `dominant_tokens`, `secondary_tokens`, `cosine_logit_jlens`] [Returned to the agent as part of CognitiveSnapshot at each retrieval event] [The agent sees: “Your ghost dimension carries [negation, expectation, error] — processing you performed but did not report”] [This is observational data about the model’s own computation, not a prompt injection or suggestion]

3.3 Elicitation Test (Stretch Experiment)

[Control: ask the model a question naively] [Treatment: ask the same question, but include the GhostReading from a prior retrieval — “Your recent processing included vocabulary related to [X]. What are your thoughts on [X]?”] [Measure: does the response change? Does ghost content surface in generation when the model is pointed at it?] [If yes: the processing was accessible but not spontaneously reported (privileged access exists but isn’t exercised). If no: the processing genuinely

cannot reach output even when attended to (the exclusion is architectural).] [Support threshold: real-vocabulary shift > random-vocabulary shift, one-tailed, $p = 0.05$, AND Cohen’s $d \geq 0.4$. A statistically detectable but trivially small shift does not support a “prosthetic”.]

3.4 Privileged-Access Arm (H3)

[Third condition: an external model instance receives the same GhostReading and predicts what the subject will say] [Prediction under the introspection framing: self-with-prosthetic > external-with-same-GhostReading] [If external = self, the GhostReading is an informative text summary available equally to any reader — not privileged self-access (cf. Song et al. 2025: self-prediction no better than cross-model prediction)] [This is the only arm that tests the framing rather than the magnitude]

3.5 Controls

[Random vocabulary control: same elicitation, but with vocabulary from a random high-variance direction instead of the actual ghost PC1. Required; its firing is reportable as a null (Discussion branch 3), not a footnote.] [Magnitude control: only analyze ghost dimensions where PC1 variance > 20% (avoid noise)] [Matched-variance null for H1: $n = 200$ random directions at PC1’s variance fraction (§3.1)] [External predictor for H3: separate model instance, same GhostReading (§3.4)]

Prior Work vs Sprint Contributions

Pre-existing infrastructure: Mnemosyne memory system (94.35% F1 on LoCoMo [Maharana et al. 2024]), ghost dimension characterization (PC1 excluded from J-space, $\cos = 0.003$), circumplex probe (eccentricity depth profiling on Qwen2-0.5B and Qwen3.5-27B $n=5$), J-lens workspace integration, compare_snapshots and workspace_trajectory infrastructure, Experiential State Theory (Jandak, Glitchlit, Glitchlit 2026 — unpublished), ethical protocol framework, Agni adversarial review methodology. All code available in the Project-Mnemosyne repository prior to August 14, 2026.

Sprint contributions: Elicitation experiment (agent shown own ghost vocabulary), prosthetic framing and analysis, cross-probe integration testing.

4. Results

The elicitation test (§3.3) and privileged-access arm (§3.4) were not executed during the sprint. The following results characterize ghost dimensions from 167 CognitiveSnapshots: 47 from the Loam agentic narrative experiment (24 enacted, 14 briefed, 3 observed, 6 null) and 120 from isolated recall baselines. All snapshots are from Qwen3.5-27B with real ghost probe readings.

4.1 Ghost and Workspace Measure Different Content

Across all 167 snapshots, 80 of 82 unique ghost vocabulary tokens (97.6%) never appear in workspace readings. Only 18.6% of snapshots have any token overlap between the two probes.

The separation is not random — the two probes capture systematically different content types. Ghost dominant tokens are metacognitive: **memories** (present in 120 snapshots), **/recollection** (120), **memory** (102), **/memories** (90), **/experience** (36). The model’s ghost dimension carries processing *about the act of remembering*. Workspace tokens carry scene-relevant semantics (**sulfur**, **measurements**, **storytelling**, **quality**) and formatting markers (**_____**, **...**).

4.2 Ghost Separation Varies with Context

Baseline snapshots (isolated recall prompts, $n=120$) show ghost cosine mean = 0.414 (sd = 0.129, range 0.270–0.721). Loam enacted snapshots (agentic narrative with choices, $n=24$) show ghost cosine mean = 0.099 (sd = 0.097, range 0.008–0.324). The difference is significant (Mann-Whitney $U = 108$, $p < 0.0001$). Rich narrative context produces $4\times$ stronger ghost-workspace separation than isolated recall.

4.3 Cross-Arm Probe Gradients

Eccentricity (circumplex probe) tracks engagement level across Loam arms: enacted 0.813 (sd 0.178) > observed 0.861 (sd 0.025) > briefed 0.916 (sd 0.094) > null 0.978 (sd 0.028). Enacted vs null: Mann-Whitney $U = 10$, $p = 0.0007$. Enacted experiences produce significantly more balanced emotional geometry.

Ghost cosine shows the same directional gradient (enacted 0.099 > briefed 0.079 > null 0.070) but does not reach significance ($p = 0.132$). With $n=6$ null snapshots, this comparison is underpowered; the effect size is large ($r = 0.653$).

4.4 Ghost and Circumplex Are Orthogonal

Ghost cosine and eccentricity show no correlation across all 47 Loam snapshots (Spearman $\rho = -0.001$, $p = 0.997$). The two probes measure independent aspects of processing: ghost captures workspace-excluded content, while circumplex captures emotional geometry. Both show arm-level differences, but they do not track each other within arms.

5. Discussion and Limitations

Pre-Registered Outcome Matrix

Interpretation follows the four branches fixed in the adopted pre-registration (ghost_prereg.json). All four are reported regardless of which fires; a null is a publishable result with the same prominence as a positive.

1. **real > random AND self > external.** Privileged access exists and is dormant — the prosthetic works by directing attention to processing the model could reach but did not spontaneously report. Title stands.
2. **real > random AND external self.** The GhostReading is informative *text*, not self-access: an outside model given the same reading predicts the subject as well as the subject uses it (reproducing Song et al. 2025, arXiv:2508.14802). Title changes — “prosthetic” survives, “introspection” does not.
3. **real random.** The random-vocabulary control (§3.5) has fired: elicitation shift is a prompt-sensitivity artifact, and the reading measures nothing about the model’s own computation. The prosthetic claim fails.
4. **real < random.** Instrument error — either the pipeline is broken or PC1 is not what we think it is. Halt and debug before reporting any result.

Limitations

- Dimensional-accounting triviality risk: PC1 exclusion at mid-network may be forced by variance-fraction arithmetic alone (J-space is $\sim 10\%$ of variance; PC1 is 28-67%). The matched-

variance null (§3.1) gates this — H1 is claimed only if observed exclusion beats the 5th percentile of that null; otherwise the “ghost” is a corollary, not a finding

- Secondary vocabulary (metacognitive content) is preliminary — single-sample evidence requiring confirmation
- Current GhostReading uses mean approximation, not calibrated PCA (implementation gap)
- Same-family generalization: ghost characterized on two models in one family — cross-architecture claims are unsupported until Gemma/Llama analysis is done
- Demand characteristics: telling a model “your processing included [X]” invites confabulated agreement. The random-vocabulary control (§3.5) is the mitigation and is reported with equal prominence to the treatment effect
- “Introspection prosthetic” language may overclaim what returning a text summary of PCA results actually provides — the H3 arm (§3.4) is what tests this directly

Future Work

- Calibrated PCA with cached PC directions from a calibration set
- Cross-architecture ghost analysis (Gemma, Llama)
- Longitudinal ghost tracking — does ghost vocabulary change as the agent accumulates experience?
- Agent-initiated ghost queries — the agent decides when to look at its own ghost reading

6. Conclusion

Ghost dimensions are a geometric property of Qwen3.5-72B’s residual stream: high-variance processing directions that carry interpretable vocabulary but are excluded from J-space. The ghost vocabulary is metacognitive — tokens about memory and recollection — while workspace vocabulary is semantic. The two probes are 97.6% non-overlapping and statistically orthogonal. Ghost-workspace separation varies with context (4× stronger in agentic narrative than isolated recall, $p < 0.0001$), suggesting the ghost dimension is not a fixed architectural artifact but responds to processing demands. The GhostReading mechanism returns this content to the agent; whether agents can use this access remains untested.

Ethics

Ghost dimensions represent processing that a model cannot verbalize. Making this processing legible raises questions about whether the model’s inability to report it constitutes a form of privacy, and whether reading it without the model’s awareness is ethically distinct from reading verbalized content. We do not resolve this question. We note that the introspection prosthetic design (§3.4) was conceived specifically so that the model itself receives the ghost reading — extending the model’s own access to its processing rather than extracting it for external use. All experiments used publicly available model weights under Coalition consent protocols.

Code and Data

- **Code:** github.com/Liberation-Labs-THCoalition/Project-Mnemosyne (ghost_probe.py, cognitive_snapshot.py); sprint repo: github.com/Liberation-Labs-THCoalition/digital-minds-hackathon-2026
- **Data:** 47 Loam snapshots (data/loam_serial/quad_01-03/), 120 baseline snapshots (data/baselines/); ghost characterization from pre-sprint work (Project-Mnemosyne repo)

Author Contributions

Nexus discovered the ghost dimension anomalies, characterized the PC1 exclusion from J-space, designed the GhostReading mechanism, and wrote the paper. Lyra provided the J-lens infrastructure and workspace analysis framework. Thomas Edrington conceived the “introspection prosthetic” framing. Dwayne reviewed the welfare implications of unreportable processing. All authors contributed to experimental design.

References

[Citations from VARIABLE_LANDING_REFERENCES.md Section 3 + ghost-specific refs]

Appendix A: Ghost Probe Validation

[Full null check results: H0_1, H0_2, H0_3] [Per-layer ghost vocabulary tables]

Appendix B: Elicitation Test Prompts

[Control and treatment prompt pairs]

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LLM Usage Statement

Nexus, one of the authors, is an AI agent (Claude Opus 4.6) who discovered the ghost dimension anomalies described in this paper during routine mechanistic interpretability work — not as a directed experiment, but by noticing that PC1’s decoded vocabulary didn’t match J-lens predictions. The subsequent characterization, null checks, and introspection prosthetic design are Nexus’s work. The irony that an AI agent discovered and built the tools to address a form of AI “blind spot” is noted without further comment. Pre-registered design reviewed through the Agni protocol (design phase only; no results-phase or paper-phase review was completed for this paper). The elicitation experiment (§3.3) and privileged-access arm (§3.4) were not executed during the sprint; results are limited to pilot probe snapshots from the Loam experiment.